

[illegible]

Pins Breakout

The diagram illustrates the pin breakout for two connectors, J4 and J3, which are 1x17 pin headers. The pins are numbered 1 through 17 on the left (J4) and 1023 through 1014 on the right (J3). The breakout is labeled '2.54 mm (0.1") breakout' on both sides. The pins are connected to various power and ground rails: 3V3, VBUS, VUSB, VIN, and VBAT. Pin 17 is connected to ground.

Pin Number	Signal
1	RST
2	I036
3	I039
4	I034
5	I035
6	I032
7	I033
8	I025
9	I026
10	I027
11	3V3
12	VBUS
13	VUSB
14	VIN
15	
16	
17	VBAT

Pin Number	Signal
1023	1
1022	2
1021	3
1019	4
1018	5
1017	6
1016	7
105	8
104	9
100	10
103	11
101	12
102	13
1015	14
1013	15
1012	16
1014	17

Power Management

Battery options (non-rechargeable):

1. Alkaline 2x1.5V or 3x1.5V (AAA, AA, B, C, D)
2. Lithium Thionyl Chloride (Li-SOCl₂) 3.6V
3. Lithium Manganese Dioxide (Li-MnO₂) 3.0V

Battery options (rechargeable):

1. Li-Ion or Li-Pol
2. 3xNiMH or 4xNiMH
3. LiFePO₄

Battery Type	R11	R13	R10	R12	Vcharge	Imin
Li-Ion/Pol	100K	10K	365K	510K	4.2V	0mA
NiMHx3	200K	10K	365K	510K	4.2V	19mA
NiMHx4	200K	10K	510K	390K	5.6V	19mA
LiFePO ₄	100K	10K	240K	510K	3.6V	0mA

Formulas:

- R11: Maintenance current (Imin)
- $Imin = ((0.44/R13) + (0.44/R11) - (VIN/R11)) \times 886$
- If Imin < 0, maintenance current is zero
- R13: Charge current = $1950/R13 = 195 \text{ mA}$
- R10, R12: Charge voltage = $2.445 \times (1 + R10/R12)$

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Battery Monitoring

The diagram shows a circuit for battery monitoring. A green line represents the I035 IC. The VBAT pin (top) is connected to a red wire labeled '1', which then goes through a 1M resistor (R7) to a green wire labeled '2'. The I035 pin labeled '2' is connected to this green wire. The I035 pin labeled '1' is connected to a red wire labeled '2', which then goes through a 1M resistor (R8) to a green wire labeled '1'. A 100nF capacitor (C7) is connected between the green wires labeled '2' and '1'. The bottom green wire is connected to a battery symbol.

Mounting

M1 M2

M2.5 size

M3 M4

The diagram shows a simple LED circuit. A yellow LED, labeled D4, is connected to a 5V power source, labeled IO2. A resistor, labeled R9 with a value of 5K1, is placed in series with the LED to limit current. The circuit is completed by a ground connection.

Notes

1. All capacitors are X7R 50V unless specified.
2. All resistors are 0.1W thick film 1% unless specified.

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